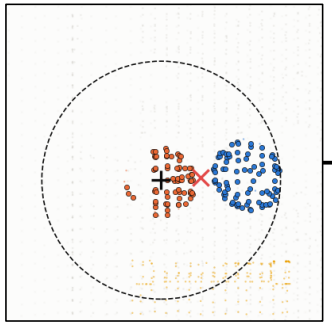
(a) Geometry-only safety filter



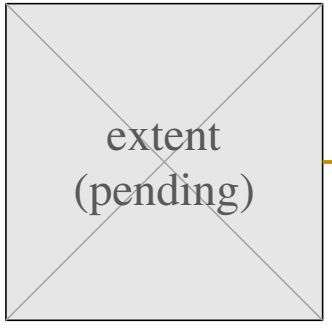
Target kept as obstacle → All links: full field → Grasp refused

(b) Target exclusion by a ball



Ball around attended cluster → All links: target-free field → Neighbor erased, contact allowed

(c) AG3S (ours)



Boundary-preserving extent → **Contact links**: target-free field → Grasp target, keep neighbor